① Processos Estocásticos (Dia 14/10/2026)

- Referência: Le-Gall (2016) Brownian Motion, Martingales and Stochastic Calculus.

Processo Gaussiano

Def: Dado $\mathcal{X} \subseteq \mathbb{R}^d$. $(f(x))_{x \in \mathcal{X}}$ é um processo Gaussiano se, pra cada $x_1, \dots, x_n \in \mathcal{X}$, $(f(x_1), \dots, f(x_n))$ é vetor gaussiano.

Dado $(f(x))_{x \in \mathcal{X}}$ processo Gaussiano, defina

$m(x) := \mathbb{E}(f(x))$, $K(x, x') := \mathbb{C}ov(f(x), f(x'))$

$\Rightarrow \forall n \in \mathbb{N}, \forall x_1, \dots, x_n \in \mathcal{X}$,

$$\begin{pmatrix} K(x_1,x_1) & \cdots & K(x_1,x_n) \\ \vdots & \ddots & \vdots \\ K(x_n,x_1) & \cdots & K(x_n,x_n) \end{pmatrix}$$

é psd. (positiva-semidefinida)

(*)

Denotamos $f \sim GP(m, K)$

Teo: Dado $m: \mathcal{X} \to \mathbb{R}$ e $\rho: \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ satisfazendo (*)

$\Rightarrow \exists f \sim GP(m, \rho)$.

Exemplo: $K(x, x') = \sigma \cdot \exp\left(-\frac{1}{2\gamma^2} \|x - x'\|_2^2 \right)$.

(2)

Movimento Browniano

Def $(B_t)_{t\geq 0}$ é movimento Browniano em $[0,\infty)$ se

(i) $B_0 = 0$

(ii) $\forall 0 \leq t_1 < \cdots < t_n$, $\{B_{t_{i+1}} - B_{t_i}\}_{i=1,\dots,n-1}$ são independentes.

(iii) $\forall 0 \leq s < t$, $B_t - B_s \sim N(0, t-s)$

(iv) $\mathbb{P}(\{\omega \in \Omega : t \mapsto B_t(\omega) \text{ é contínuo}\}) = 1$

$(B_t)_{t\in[0,T]}$ é movimento Browniano em $[0,T]$.

- $(B_t)_{t\geq 0}$ movimento Browniano $\Rightarrow (B_t)_{t\geq 0} \sim GP(0, K)$, onde

$\forall s \geq t$:

$$K(s,t) = \mathbb{E}(B_s B_t) = \mathbb{E}\left((B_s - B_t)B_t + B_t^2\right)$$
$$= \mathbb{E}((B_s - B_t))\mathbb{E}(B_t) + t$$
$$= t$$

$\forall s \leq t : K(s,t) = s$

$$\Rightarrow \quad K(s,t) = s \wedge t$$

Teo: (Propriedade de Markov) $(B_{t+s} - B_s)_{t\geq 0}$ é "independente de" $\mathcal{F}_s = \sigma(B_u : u \leq s)$

Teo: (Propriedade Forte de Markov): τ tempo de parada, $(B_t)_{t\geq 0}$ movimento Browniano, $\mathcal{F}_t^B = \sigma(B_u : u \leq t)$.

$\Rightarrow (B_{\tau+t} - B_\tau)_{t\geq 0}$ é MB independente de $\mathcal{F}_\tau^B$.

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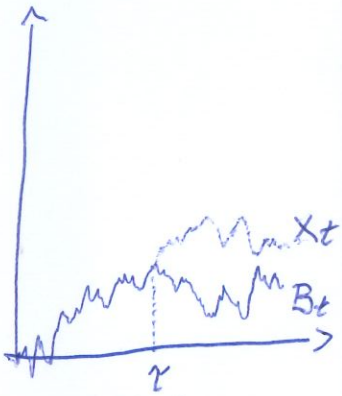
Teo: $(B_t)_{t\geq 0}$ é martingal c.r.a $(\mathcal{F}_t)_{t\geq 0}$

prova: $\mathbb{E}(B_t \mid \mathcal{F}_s) = \mathbb{E}(B_t - B_s \mid \mathcal{F}_s) + B_s = B_s$ □

Teo (Princípio da Reflexão) τ tempo de parada c.r.a $(\mathcal{F}_t^B)_{t\geq 0}$.

$$X_t = \begin{cases} B_t, & \text{se } t < \tau \\ B_\tau - (B_t - B_\tau), & \text{se } t \geq \tau \end{cases}$$

$\Rightarrow$ $(X_t)_{t\geq 0}$ é Movimento Browniano.



Teo: $B_t^* = \max_{s\in[0,t]} B_s$.

$$\Rightarrow \mathbb{P}\left(B_t^* \geq a, B_t \leq b\right) = \mathbb{P}\left(B_t \geq 2a - b\right) \quad \forall a \geq b, \; a \geq 0, \; t > 0.$$

$\Rightarrow B_t^* \sim |B_t|$.

prova:

$\tau_a = \inf\{t \geq 0 : B_t = a\}$. $(\tau_a < \infty \text{ a.s})$

$\Rightarrow \mathbb{P}(B_t^* \geq a, B_t \leq b) = \mathbb{P}(\tau_a \leq t, B_t \leq b)$

$= \mathbb{P}(\tau_a \leq t, \tilde{B}_{t-\tau_a} + B_{\tau_a} \leq b)$ onde $\tilde{B}_t = B_{t+\tau_a} - B_{\tau_a}$.

$= \mathbb{P}(\tau_a \leq t, \tilde{B}_{t-\tau_a} \leq b - a)$ (pois $B_{\tau_a} = a$) (*)

(4)

Markov Forte

~~$\Rightarrow \mathbb{P}(\tau_a < t, \tilde{B}_{t-\tau_a} \leq b-a) = \mathbb{P}(\tau_a < t) \mathbb{P}(\tilde{B}_{t-\tau_a} \leq b-a)$~~

Seja $H = \{(s,w) \in \mathbb{R}_+ \times C(\mathbb{R}_+, \mathbb{R}) : s \leq t,\ w(t-s) \leq b-a\}$

$(*) \Rightarrow \mathbb{P}(B_t^* \geq a, B_t \leq b) = \mathbb{P}((\tau_a, \tilde{B}) \in H)$.

Markov Forte + Simetria do MB $\Rightarrow (\tau_a, \tilde{B}) \sim (\tau_a, -\tilde{B})$.

$\Rightarrow \mathbb{P}(B_t^* \geq a, B_t \leq b) \overset{(*)}{=} \mathbb{P}((\tau_a, \tilde{B}) \in H)$

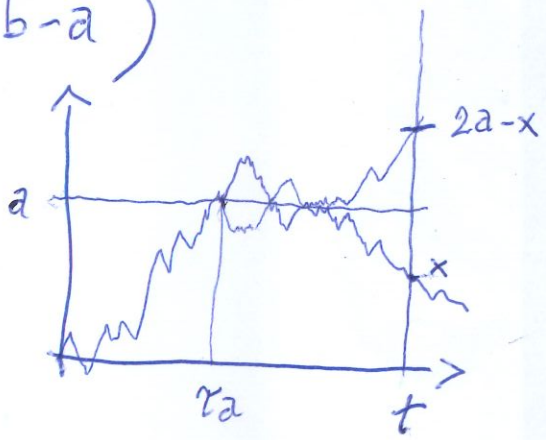
$= \mathbb{P}((\tau_a, -\tilde{B}) \in H)$

$= \mathbb{P}(\tau_a \leq t, -\tilde{B}_{t-\tau_a} \leq b-a)$

$= \mathbb{P}(\tau_a \leq t, -B_t + B_{\tau_a} \leq b-a)$

$= \mathbb{P}(\tau_a \leq t, B_t \geq 2a-b)$

$= \mathbb{P}(B_t \geq 2a-b)$ ∎



$\mathbb{P}(B_t^* \geq a) = \mathbb{P}(B_t^* \geq a, B_t \geq a) + \mathbb{P}(B_t^* \geq a, B_t < a)$

$= \mathbb{P}(B_t \geq a) + \mathbb{P}(B_t \geq a)$

$= \mathbb{P}(B_t \geq a) + \mathbb{P}(-B_t \leq -a)$

$= \mathbb{P}(|B_t| \geq a)$. ∎

⑤ Prop: $(B_t)_{t\geq 0}$ MB. Então também são MBs os seguintes:

(i) $X_t = \frac{1}{\sqrt{a}} B_{at}$, $a>0$

(ii) $X_t = t B_{1/t}$, $t>0$, $X_0 = 0$.

(iii) $X_t = B_{t+s} - B_s$, $t \geq 0$

(iv) $X_t = B_T - B_{T-t}$, $0 \leq t \leq T$.

(v) $X_t = -B_t$.

Existência do Movimento Browniano

Seja $\mathcal{H}$ espaço de Hilbert com produto interno $\langle \cdot, \cdot \rangle$, norma $\|\cdot\|$, base ortonormal enumerável $(\varphi_n)_{n\in\mathbb{N}} \in \mathcal{H}$.

$$\Rightarrow \forall f \in \mathcal{H}, \exists (a_n)_{n\in\mathbb{N}} : f = \sum_{n=1}^{\infty} a_n \varphi_n .$$

- $a_n = \langle f, \varphi_n \rangle$
- $\|f\|_2^2 = \sum_{n=1}^{\infty} \langle f, \varphi_n \rangle^2$
- Dado $g = \sum_{n=1}^{\infty} b_n \varphi_n$, vale $\langle f, g \rangle = \sum_{n=1}^{\infty} a_n b_n$

Tome $\mathcal{H} = L^2[0,1]$, $\langle f, g \rangle = \int_0^1 f(x) g(x) dx$.

- Seja $(Z_n)_{n\in\mathbb{N}}$ iid $N(0,1)$
- Defina $X_t = \sum Z_n \langle \mathbb{1}_{[0,t]}, \varphi_n \rangle = \sum Z_n \int_0^t \varphi_n(x) dx$